

HIV, AIDS and Human Retroviruses

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Session Objectives

1. Explain the origins of HIV
2. Describe the classification, structure and life cycle of HIV-1
3. Identify the mechanism of HIV transmission and pathogenesis
4. Describe the untreated HIV timeline
5. Compare therapeutic options and mechanisms of action
6. Manage newly diagnosed HIV infection

Recommended Reading:

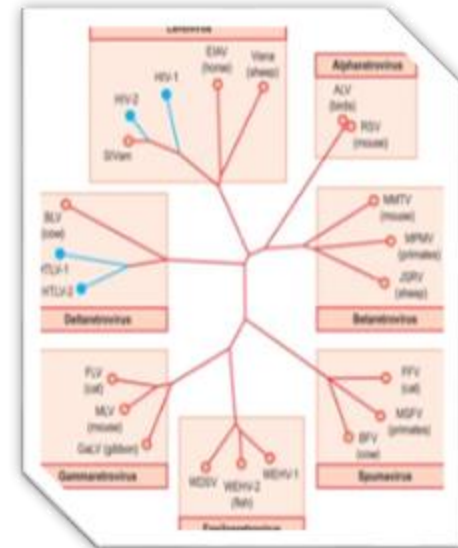
[HTLV: Section on Human T-Cell Lymphotropic Virus in Harrisons](#)

Question 1

What oncogenic virus causes ~500,000 cases of lymphoma or leukemia each year?

1. Hepatitis C virus
2. Epstein-Barr virus
3. HTLV-1
4. HIV

- HTLV-1: adult T-cell leukemia/lymphoma & HTLV-1 Associated Myelopathy (Tropical Spastic Paraparesis)
- Trophic to mature T-cells & GLUT-1 receptor
- 5-20 million infected worldwide with clusters (Japan, Caribbean, Central/South America, & Africa)
- ATLL: cutaneous lesions, lytic bone lesions, hypercalcemia
- Poor prognosis (5-yr survival <5%)
- Transmission:
 - a) **Breastfeeding (important mode)**
 - b) Transfusion
 - c) Contaminated needles
 - d) Sexual intercourse

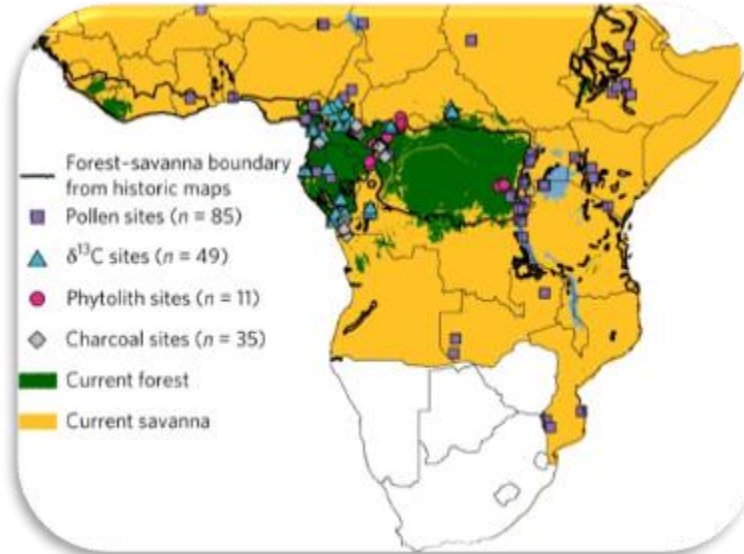


- Bobby, 25-year-old male comes for STI testing at the after unprotected sex.
 - A complete history and physical examination, including anogenital exam, is unremarkable
 - You recommend tests for HIV, syphilis, gonorrhea, Chlamydia, HBV and trichomonas.
 - Despite unprotected exposures and the USPSTF recommendation that all between the ages of 15 and 65 have at least one HIV test and, he has refused HIV testing previously because he believes HIV is a government plot.
- Where do you tell Bobby HIV came from?

The Origin of HIV

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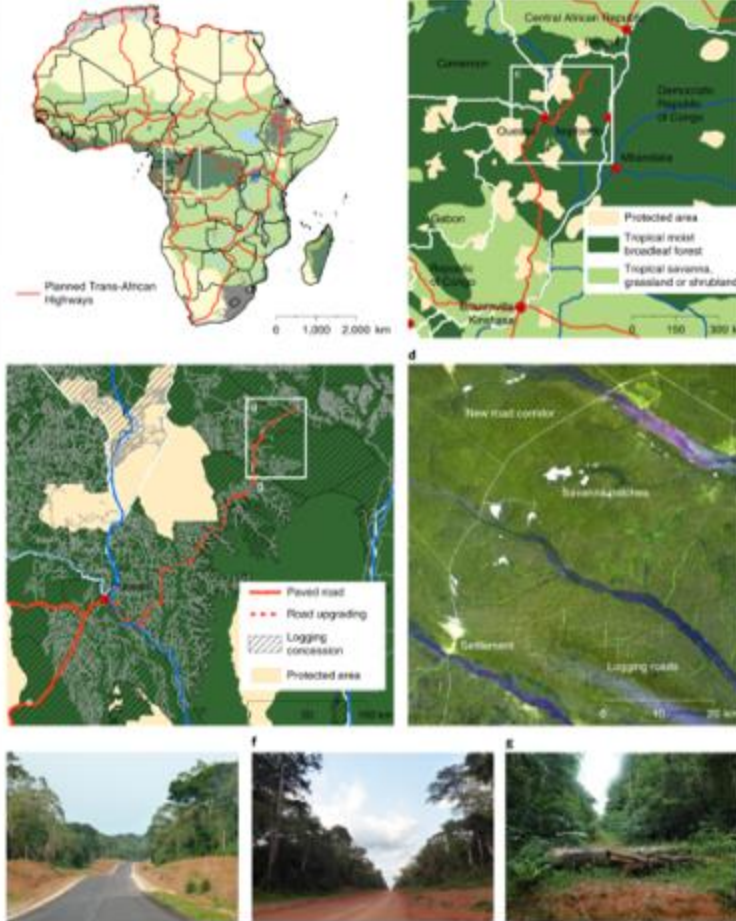
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Detection and Partial Characterization of Simian Immunodeficiency Virus SIVsm Strains from Bush Meat Samples from Rural Sierra Leone

Cristian Apetrei^{1,2,*} Michael J. Metzger¹ David Richardson³ Binhua Ling¹ Paul T. Telfer¹ Patricia Reed¹ David L. Robertson³ and Preston A. Marx^{1,2}

Journal of
Virology*



SIV

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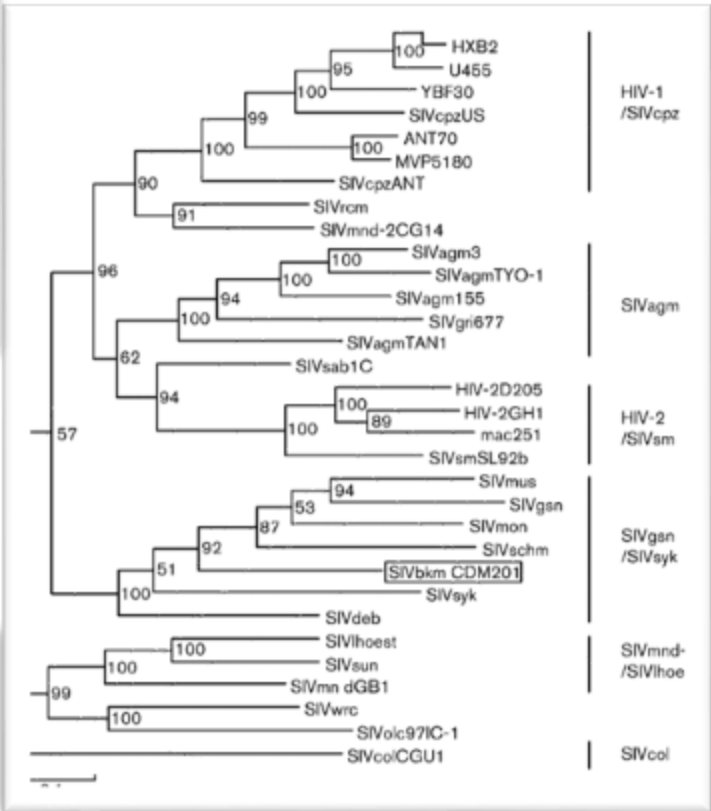
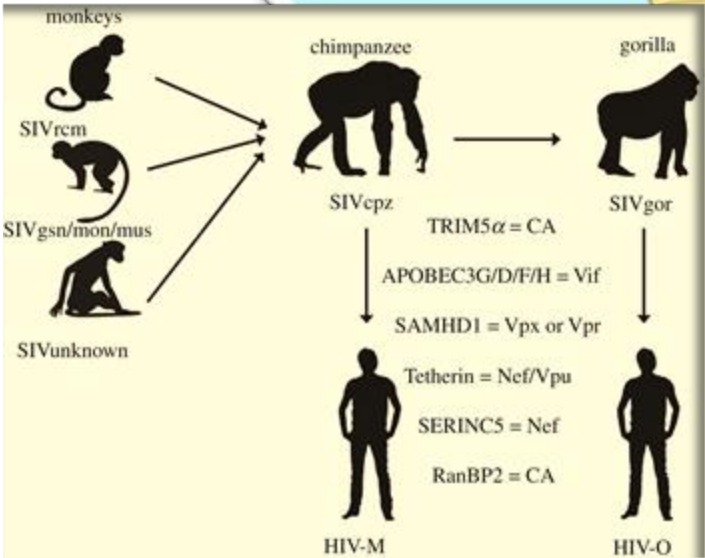
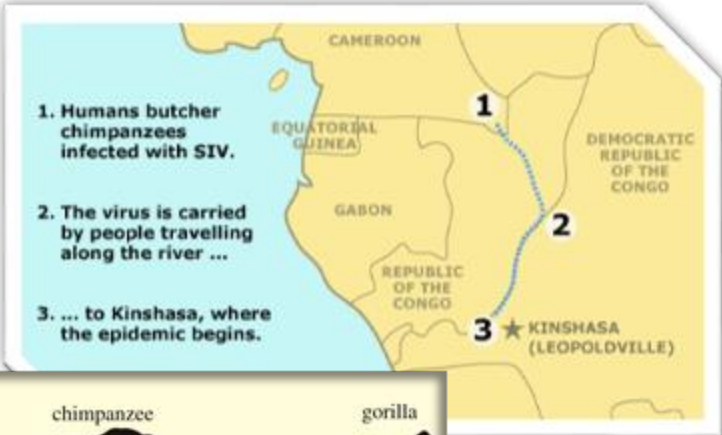
Question 2

What virus was transmitted to humans that eventually evolved into HIV-1 clade M?

1. SIVcpz
2. SIVgor
3. SIVgsn/mon/mus
4. SIVrcm



SIV Evolves to Produce a New Virus: HIV



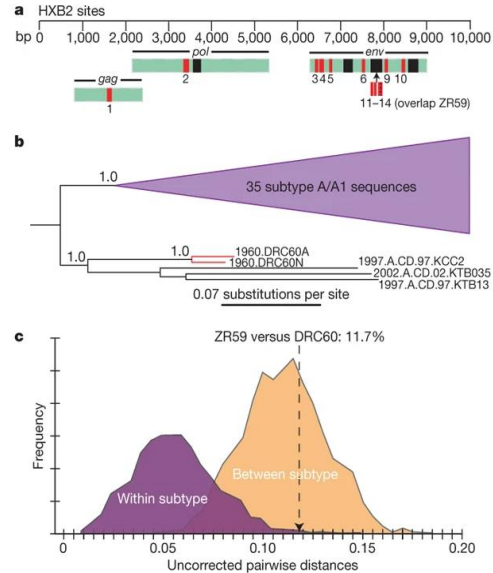
Question 3

In what year is the first known documented case of HIV infection?

1. 1884
2. 1924
3. 1959
4. 1981

Direct evidence of extensive diversity of HIV-1 in Kinshasa by 1960

Michael Worobey¹, Marlea Gemmel¹, Dirk E. Teuwen^{2,3}, Tamara Haselkorn¹, Kevin Kunstman⁴, Michael Bunce⁵, Jean-Jacques Muyembe^{6,7}, Jean-Marie M. Kabongo⁶, Raphaël M. Kalengayi⁶, Eric Van Marck⁸, M. Thomas P. Gilbert¹† & Steven M. Wolinsky⁴



Earliest Documented Case of HIV: 1959

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FATAL INCLUSION-DISEASE PNEUMONITIS IN AN ADULT *

GARDNER C. McMILLAN, M.D. †

(From the Department of Pathology, Pathological Institute, McGill University, Montreal, Que.)

While occasional cases of inclusion disease affecting various organs in infants have been reported for many years, the condition remains distinctly rare in adults. The following case represents, so far as I am aware, the ninth to be reported in an adult, and the second example of adult inclusion-disease pneumonitis. [Am J Pathol.](#) 1947 Nov; 23(6): 995-1003.

PEDIATRICS®

OFFICIAL JOURNAL OF THE AMERICAN ACADEMY OF PEDIATRICS

Armond S. Goldman, Lynn R. Goldman and Daniel A. Goldman

Pediatrics June 2005, 115 (6) e725-e736; DOI: <https://doi.org/10.1542/peds.2004-2157>

What Caused the Epidemic of *Pneumocystis* Pneumonia in European Premature Infants in the Mid-20th Century?

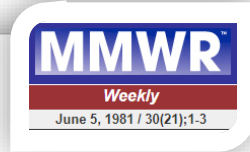


> MMWR Morb Mortal Wkly Rep. 1981 Jul 3;30(25):305-8.

Kaposi's sarcoma and Pneumocystis pneumonia among homosexual men--New York City and California

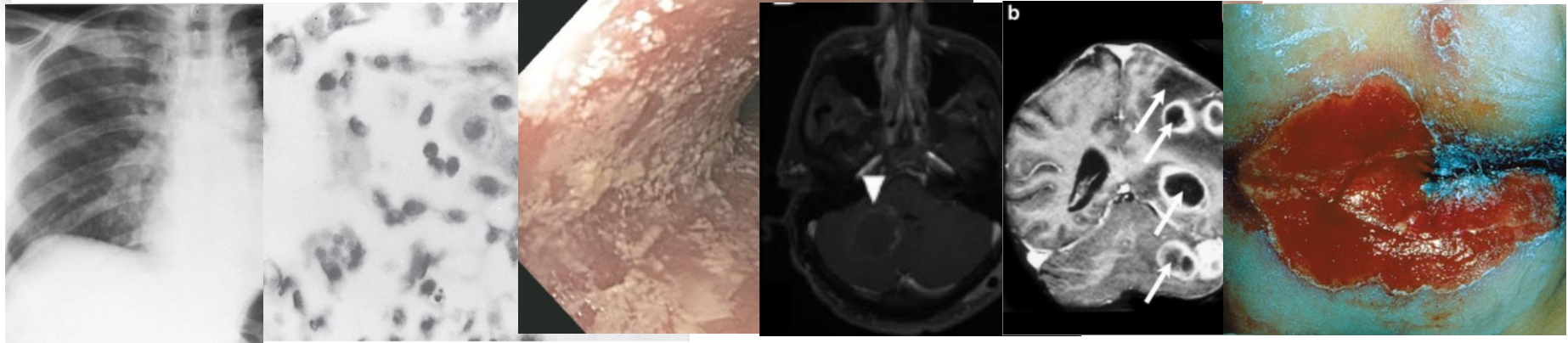
Pneumocystis Pneumonia --- Los Angeles

In the period October 1980-May 1981, 5 young men, all active homosexuals, were treated for biopsy-confirmed *Pneumocystis carinii* pneumonia at 3 different hospitals in Los Angeles, California. Two of the patients died. All 5 patients had laboratory-confirmed previous or current cytomegalovirus (CMV) infection and candidal mucosal infection. Case reports of these patients follow.



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Devastation

Discovery and Action

End of AIDS

1980

1985

1990

1995

2000

2005

2010

2015

2020

1981 AIDS
1983 HIV
1983 WHO surveillance
1983 Denver Principles
1984 laboratory reagents and early blood test

1985 Commercial HIV test
1985: early AZT studies
1987 FDA-approved AZT
1987 WHO GPA
1987 TASO Uganda
1985 Reagan mentions AIDS

1992 AIDS #1 killer US men 25-44 years old
1993 US Office of National AIDS Policy
1994 AZT to prevent MTC

1995 Protease inhibitor
1996 Vancouver triple therapy
1996 US home HIV test
1997 AIDS deaths decline 40% in US
1998 TAC South Africa

2001 Special UN Session "global emergency"
2002 Leading cause of death
2002 Global Fund established
2003 PEPFAR
2003 WHO 3x5

2005 1st generic ARVs
2006 TasP proposed as HIV control strategy
2008 Swiss statement
2009 Attia meta-analysis ART prevents transmission
2009 WHO proposes universal testing and treatment to eliminate HIV in South Africa

2011 HPTN 052 proves ART blocks transmission
2012 PrEP approved in US
2010 UNAIDS Treatment 2.0 with treatment as prevention
2014 UNAIDS 90-90-90 targets

2015 Fast Track Cities with 60+ cities
2015 START & TEMPRANO
2015 Access to treatment is human right
2016 18.2 M (49%) people on treatment
2016 Since start of epidemic 39 million AIDS deaths

2020 90-90-90 reached
2030 95-95-95 targets affirmed

2020 40 M PLHIV
32M (81%) Tx

1980
100,000 PLHIV
No treatment

1985
Nearly million PLHIV
No treatment

1990
Millions PLHIV
No treatment

1995
20M PLHIV
No Tx

2000
28.9M PLHIV
<200,000 (1%) Tx
Africa
<50,000 (1%)

2005
31.8M PLHIV
1.3M (4%) Tx

2010
33.3M PLHIV
7.5M (23%) treatment

2015
36.7M PLHIV
17M (46%) Tx



Policy

Hit early, hit hard. Almost no access to treatment in low and middle income countries.

2003 WHO CD4 <200

2009 WHO CD4 <350

2012 US Treat All
2013 WHO CD4 <500

2015 WHO Irrespective CD4

Devastation

Discovery and Action

End of AIDS

1980

1985

1990

1995

2000

2005

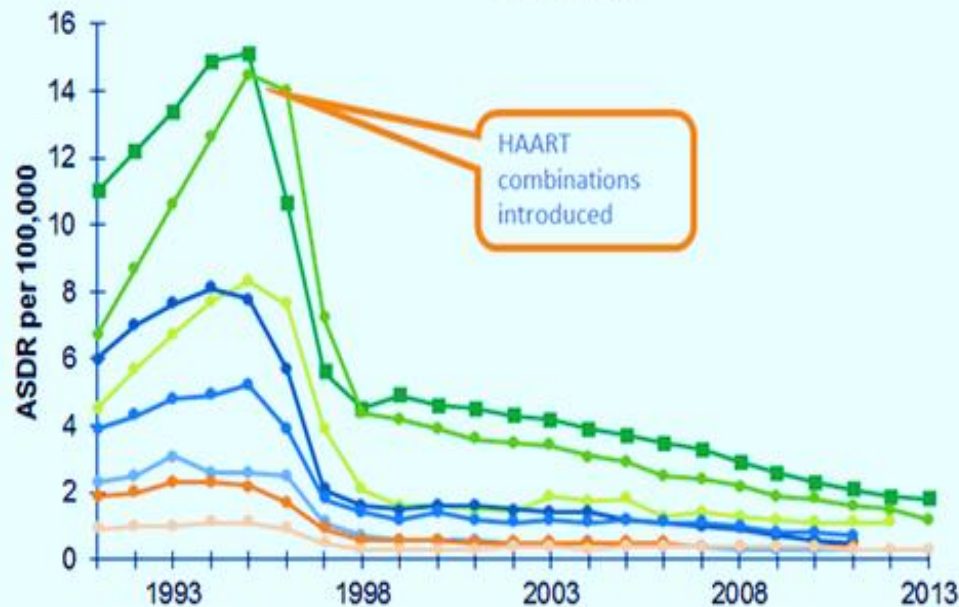
2010

2015

2020

HIV/AIDS Age-Standardized Death Rates (ASDR)

By Country



Country	Decline in ASDR (1995-2013*)
USA	-88%
Spain	-92%
Italy	-87%
France	-94%
Canada	-87%
Australia	-88%
Germany	-82%
United Kingdom	-73%

2015 Fast Track Cities with 60+ cities
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Devastation

Discovery and Action

End of AIDS

1980

1985

1990

1995

2000

2005

2010

2015

2020

1981 AIDS cases reported
1983 HIV discovered
1983 WHO surveillance system
1983 IDU as principle risk factor
1984 labor unions reage early test

AIDS AND ACCUSATION

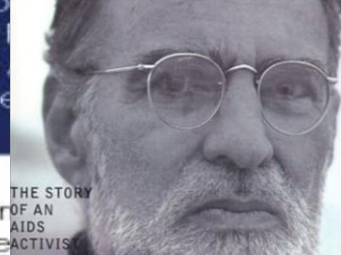
HAITI AND THE GEOGRAPHY OF BLAME

PAUL FARMER

1992 AIDS #1 killer US men 25-44 years
1993 Office of National AIDS prevention

1990 100,000 PLHIV
No treatment

REPORTS FROM THE HOLOCAUST



LARRY KRAMER

UPDATED AND EXPANDED
FOREWORD BY SIMON WATNEY

1995 Protease inhibitors
2001 Special UN Session on global emergency
2002 Leading cause of death
2002 Global Fund established
2003 PEPFAR
2003 WHO 3x5

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2002 Leading cause of death
2002 Global Fund established
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2000 28.9M PLHIV
<200,000 (1%) Tx
Africa
<50,000 (1%)

2005 1st generic ARVs
2006 TasP proposed as

THE CLASSIC BESTSELLER
"STUNNING...AN IMPRESSIVELY RESEARCHED AND RICHLY DETAILED NARRATIVE."—TIME

AND POLITICS, PEOPLE, AND THE AIDS EPIDEMIC

THE BAND PLAYED ON

RANDY SHILTS

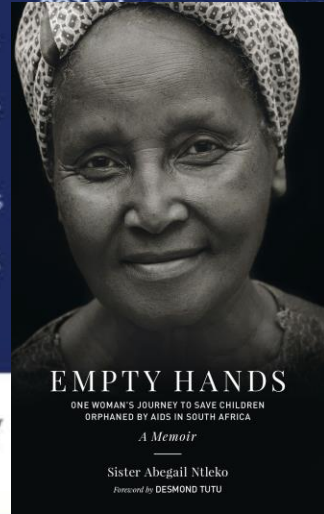
2005 31.8M PLHIV
1.3M (4%) Tx

2011 HPTN 052 proves ART blocks transmission
2015 2 PrEP proved in

2010 10 UNAIDS treatment with
2011 4 UNAIDS 0-90
2012 0-90
2013 0-90
2014 0-90
2015 0-90
2016 0-90
2017 0-90
2018 0-90
2019 0-90
2020 0-90

2010 3M PLHIV
1M (23%) treatment

2015 Fast Track Cities with 60+ cities
2015 START & TEMPRANO



EMPTY HANDS

ONE WOMAN'S JOURNEY TO SAVE CHILDREN ORPHANED BY AIDS IN SOUTH AFRICA

A Memoir

Sister Abegail Ntleko

Foreword by DESMOND TUTU

2020 90-90-90 reached
2030 95% reached

95% reached

PLHIV (1%) Tx

Policy

Hit early, hit hard. Almost no access to treatment in low and middle income countries

2003 WHO CD4 <200

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2012 US Treat All
2013 WHO CD4 <500

2015 WHO Irrespective CD4

Bobby

- Bobby reported unprotected sex with multiple women and men, has been treated for multiple STIs including gonorrhea and syphilis and has recurrent genital herpes. He shared needles with others when using heroin, last 12 years ago.
- What do you estimate is Bobby's risk of acquiring HIV infection?

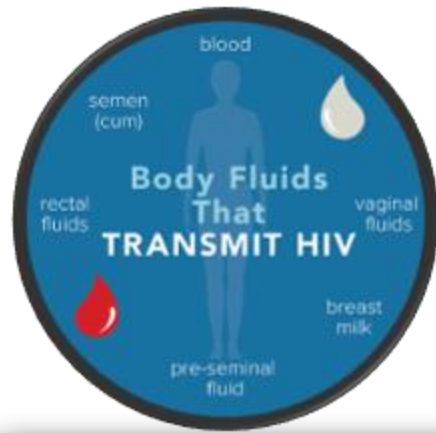
Question 4

In the U.S., of the following what is the most common mode of HIV transmission?

1. Sharing utensils
2. Kissing
3. Mother-to-child
4. Injection drug use

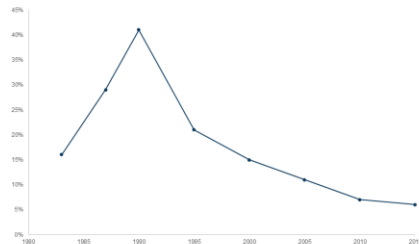
HIV Routes of Transmission

In the presence of genital ulcerative STIs the risk of HIV transmission may increase up to **5x**



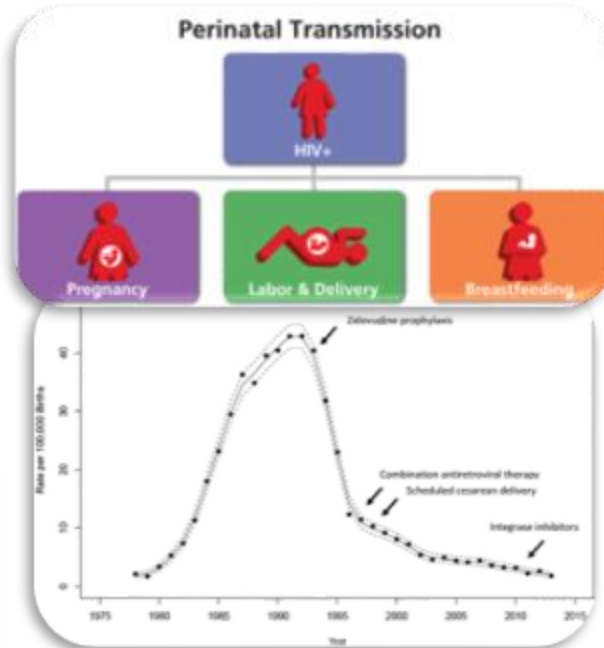
Sharing needles, syringes, or other drug injection equipment—for example, cookers—puts people at risk for getting or transmitting HIV and other infections.

Figure 1
Share of HIV Infections Attributed to Injection Drug Use Overtime, 1980-2015



Source: Data through 2006 - CDC data request Data from 2008-2015 - CDC Atlas
Note: Methodologies used to obtain data from the two different sources may differ and may not directly compare

KFF



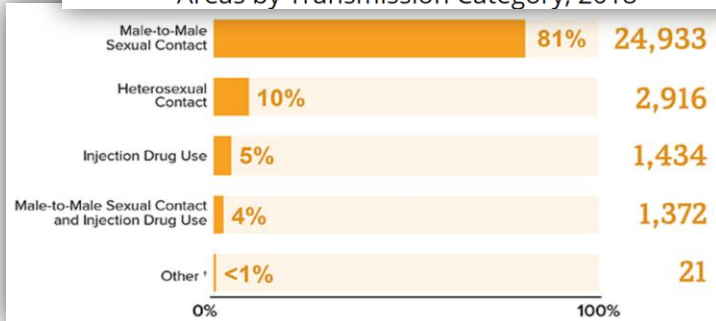
Question 5

Compared to proportion of population, what group has the highest incidence of HIV infection?

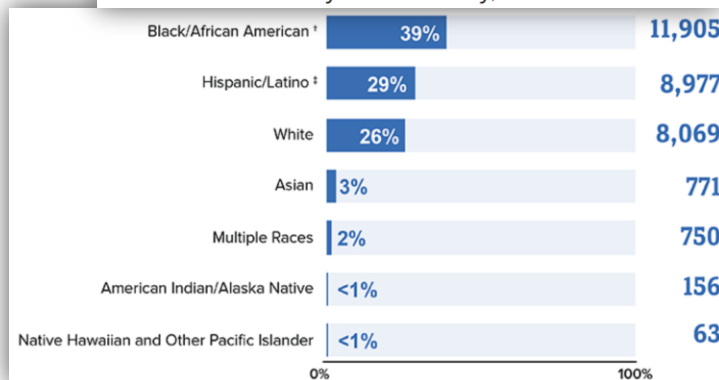
1. Latin females
2. Black males
3. White females
4. White males

HIV Diagnoses in Men

New HIV Diagnoses Among Men in the US and Dependent Areas by Transmission Category, 2018*



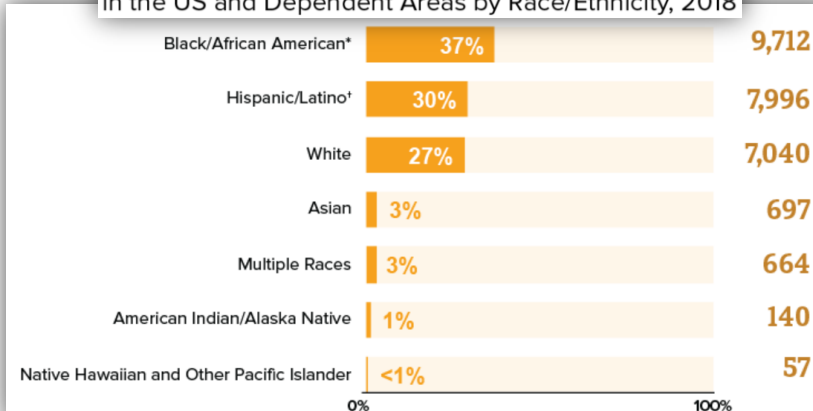
New HIV Diagnoses Among Men in the US and Dependent Areas by Race/Ethnicity, 2018*



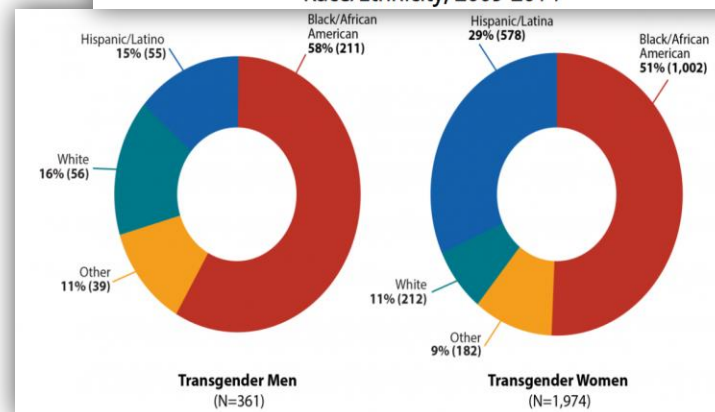
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New HIV Diagnoses Among Gay and Bisexual Men in the US and Dependent Areas by Race/Ethnicity, 2018



HIV Diagnoses Among Transgender People in the United States^c by Race/Ethnicity, 2009-2014



Discussion 1

Incidence of HIV is highest in the southeast U.S.
Why do you think that is?

HIV Diagnoses in Women

New HIV Diagnoses Among Women by Transmission Category in the US and Dependent Areas, 2018*

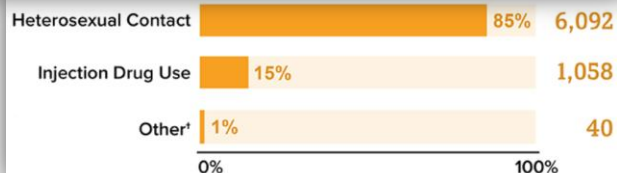


Figure 20. Rates of Diagnoses of HIV Infection among Female Adults and Adolescents, 2018—United States and 6 Dependent Areas

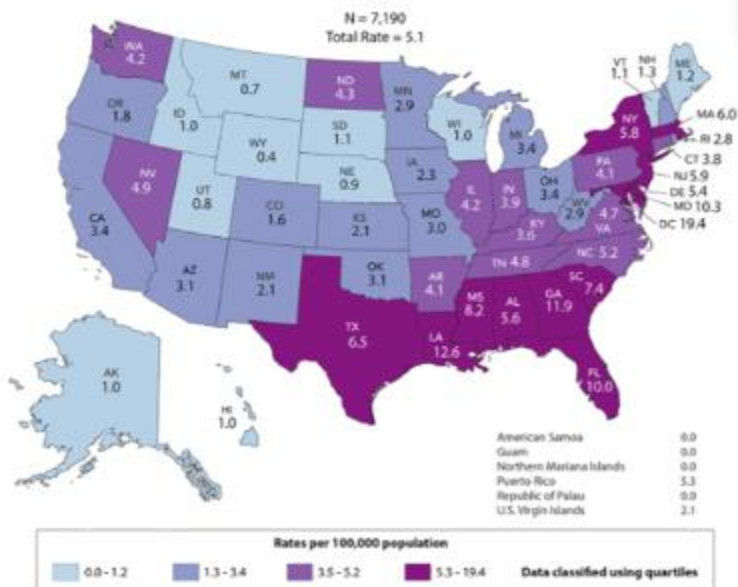


Figure 23. Percentages of Diagnoses of HIV Infection among Female Adults and Adolescents, by Race/Ethnicity and Transmission Category, 2018—United States and 6 Dependent Areas

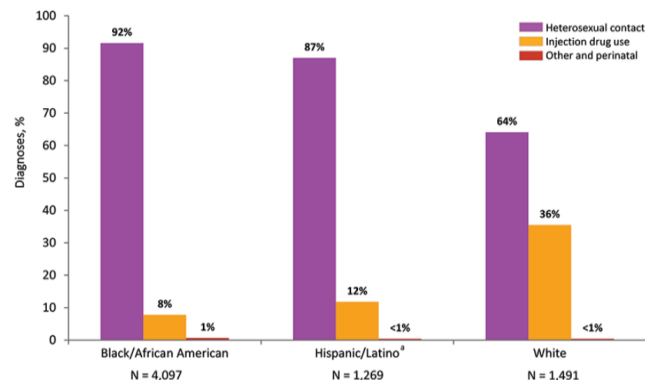
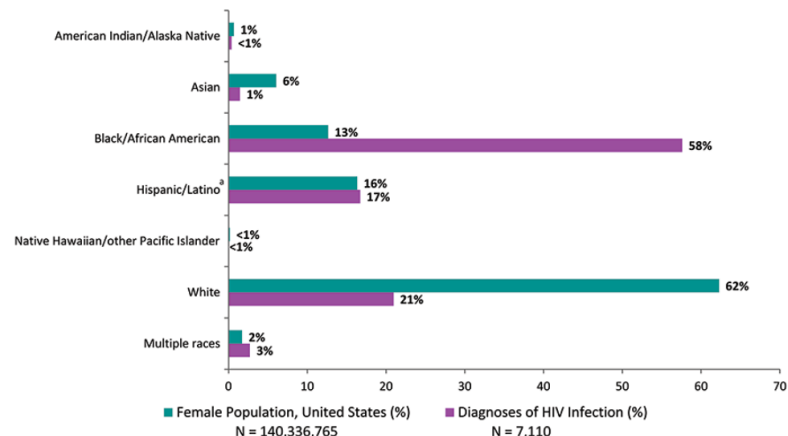
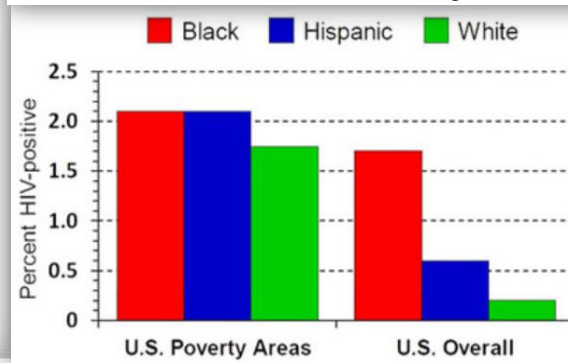


Figure 21. Percentages of Diagnoses of HIV Infection and Population among Female Adults and Adolescents, by Race/Ethnicity, 2018—United States



HIV: Poverty, Geographics

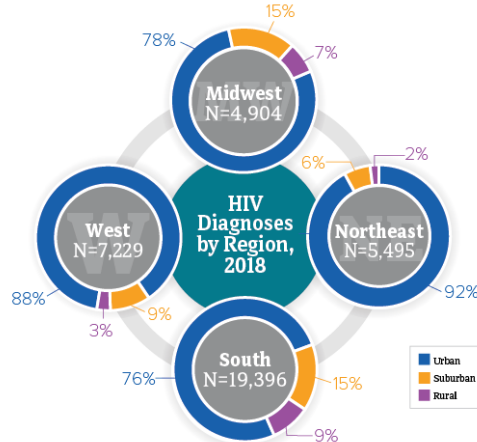
HIV Prevalence Rate, by Race/Ethnicity



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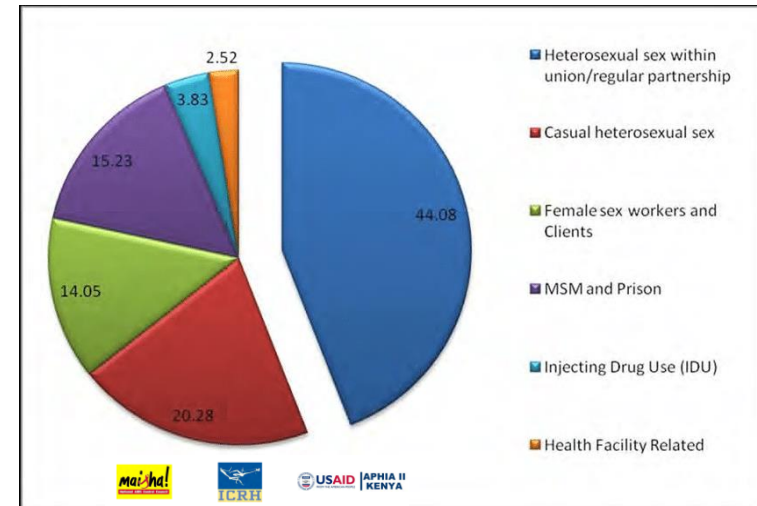
HIV Diagnoses in the 50 States and the District of Columbia by Region, 2018*



*Data excludes persons whose county of residence is unknown.

Source: CDC. HIV surveillance in urban and nonurban areas through 2018 (slide)

Findings from a Three-Site PLACE Study in Kenya



mai-ha!

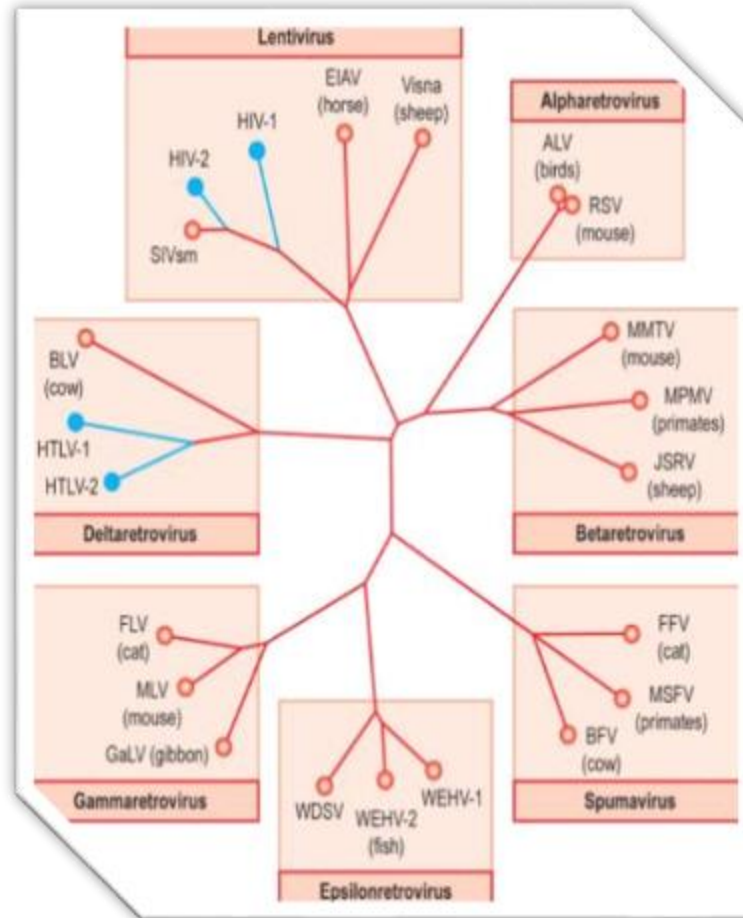
ICRH

USAID APHIA II KENYA

- You tell Bobby that his behavior puts him at very high risk of getting HIV and recommend he have HIV testing. He admits he doesn't understand what a virus is and how it can infect him and make him sick.
- What can you tell Bobby to make him more informed?

Family: Retroviridae

- Human pathogens
 - HIV-1
 - HIV-2
 - HTLV-1
 - HTLV-2
 - HTLV-3
 - HTLV-4



Question 6

The HIV genome contains two copies of

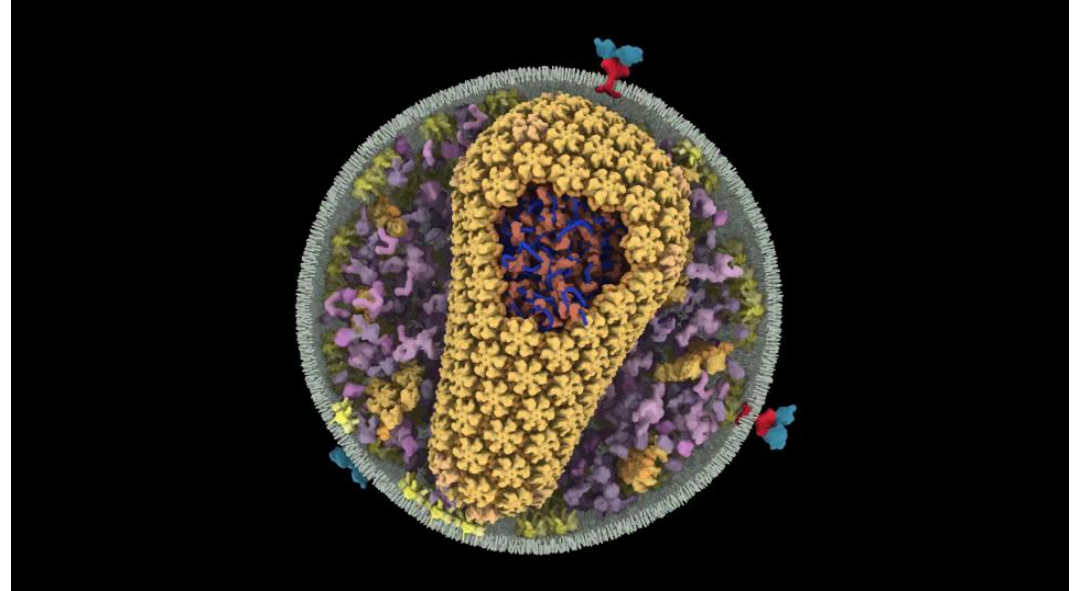
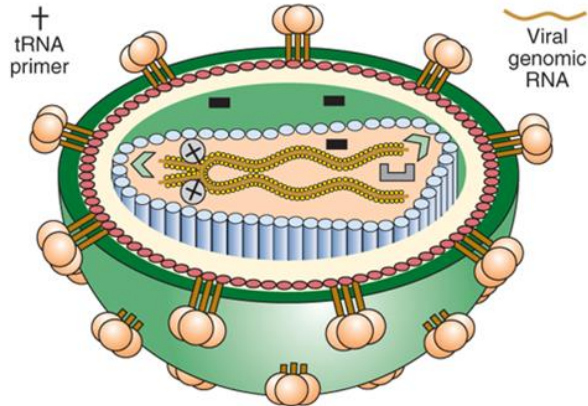
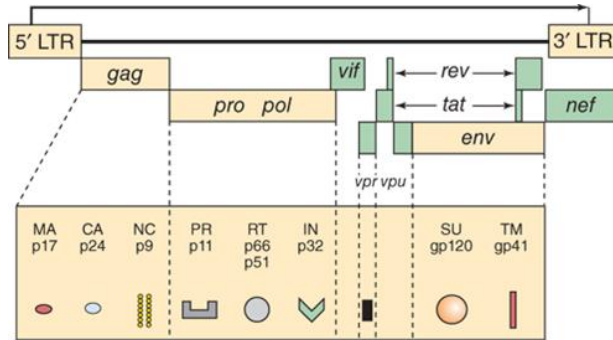
1. ssDNA
2. dsDNA
3. ssRNA
4. dsRNA

Question 7

Match the HIV genome reading frame with the protein that it encodes.

- | | |
|----------|-----------|
| 1. gp120 | A. 3'-LTR |
| 2. nef | B. gag |
| 3. p17 | C. env |
| 4. p66 | D. 5'-LTR |
| 5. gp41 | E. pol |
| 6. gp160 | F. rev |
| 7. rev | G. IN |
| 8. IN | H. Vif |

HIV: Structure



[Link to full video.](#)

Question 8

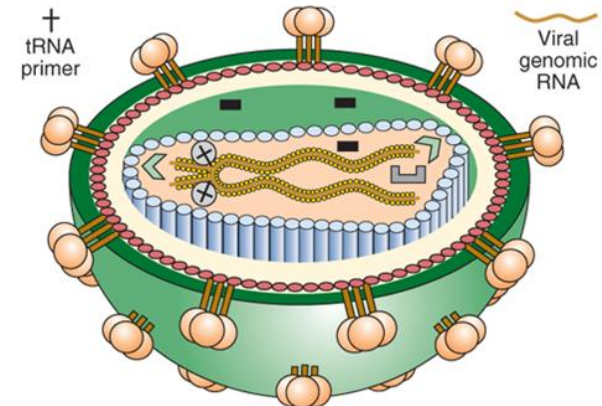
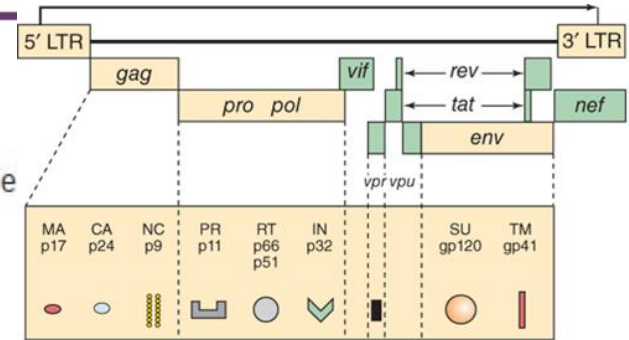
Match the HIV protein with its function

- | | |
|-------------|--|
| 1. Nef | A. Capsid protein |
| 2. gp120 | B. Cleaves gag, pol precursors |
| 3. p17 | C. Fusion of virion and T-helper cell membrane |
| 4. p24 | D. Export of viral RNA from nucleus |
| 5. Vif | E. Degrades host cell defense proteins |
| 6. gp160 | F. Downregulates host CD4, MHC-1 |
| 7. Rev | G. Binds CD4, CCR5, CXCR4 |
| 8. Protease | H. Matrix protein |

What are the major proteins encoded by the HIV genome?



Gene	Protein (s)	Major function (s)
Gag	p24, p17 p7, p6	Capsid protein, matrix protein Capsid protein, capsid protein
Pol	Reverse transcriptase Integrase Protease	Transcribes viral RNA into dsDNA Integrates viral dsDNA into host genome Cleaves gag and pol precursors
Vif	Vif	Degrades host cell defense protein APOBEC3G
Vpr	Vpr	Regulates nuclear import of pre-integration complex
Tat	Tat	Promotes transcription of viral DNA
Rev	Rev	Allows export of unspliced viral RNA from nucleus
Vpu	Vpu	Intracellular degradation of CD4
Env	gp120 gp41	Binds CD4, CCR5, and CXCR4 Promotes fusion of virion with plasma membrane
Nef	Nef	Downregulates of Cluster of differentiation 4, MHC-1 and other receptors



Source: S. Riedel, J.A. Hobden, S. Miller, S.A. Morse, T.A. Mietzner, B. Detrick, T.G. Mitchell, J.A. Sakanari, P. Hotez, R. Mejia Jawetz, Melnick, & Adelberg's Medical Microbiology, 28e Copyright © McGraw-Hill Education. All rights reserved.

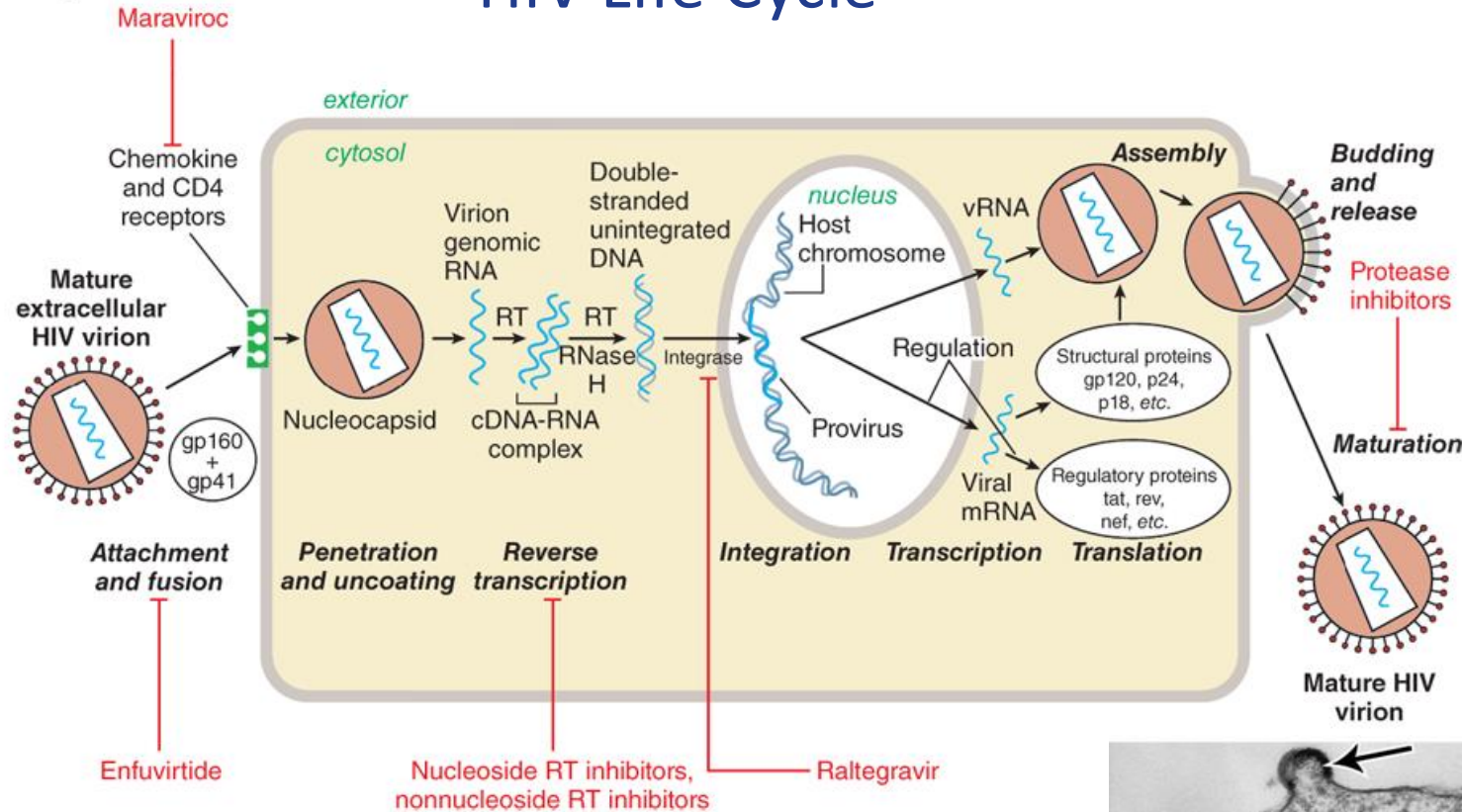
[Link to full video.](#)

Question 9

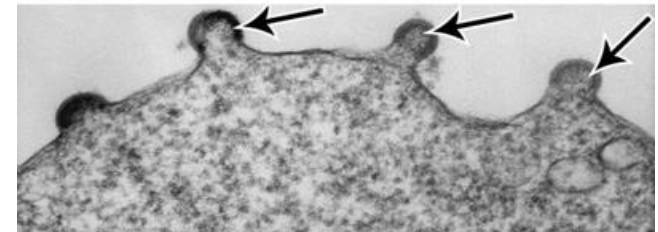
Match the ART with its mechanism of action

- | | |
|-----------------|---|
| 1. Maraviroc | A. False nucleoside for reverse transcriptase |
| 2. Dolutegravir | B. Inhibits action of gp41 |
| 3. Enfuvirtide | C. Inhibits reverse transcriptase and integrase |
| 4. Tenofovir | D. Prevents viral maturation |
| 5. Rilpivirine | E. Prevents formation of provirus |
| 6. Atazanavir | F. Prevents transcription of viral RNA |
| 7. Emcitribine | G. Binds CXCR4 |
| 8. Biktarvy® | H. Binds CCR5 |

HIV Life Cycle



[Link to full video.](#)



Question 10

The test for maraviroc activity is a

1. HIV-1 genotype
2. HIV-1 phenotype
3. HIV-1 integrase genotype
4. Coreceptor tropism assay

Question 11

The average time from infection until symptomatic HIV infection is

1. 3-5 years
2. 6-7 years
3. 8-10 years
4. 12-13 years

Discussion 2

What is a CD4 count and viral load?

Question 12

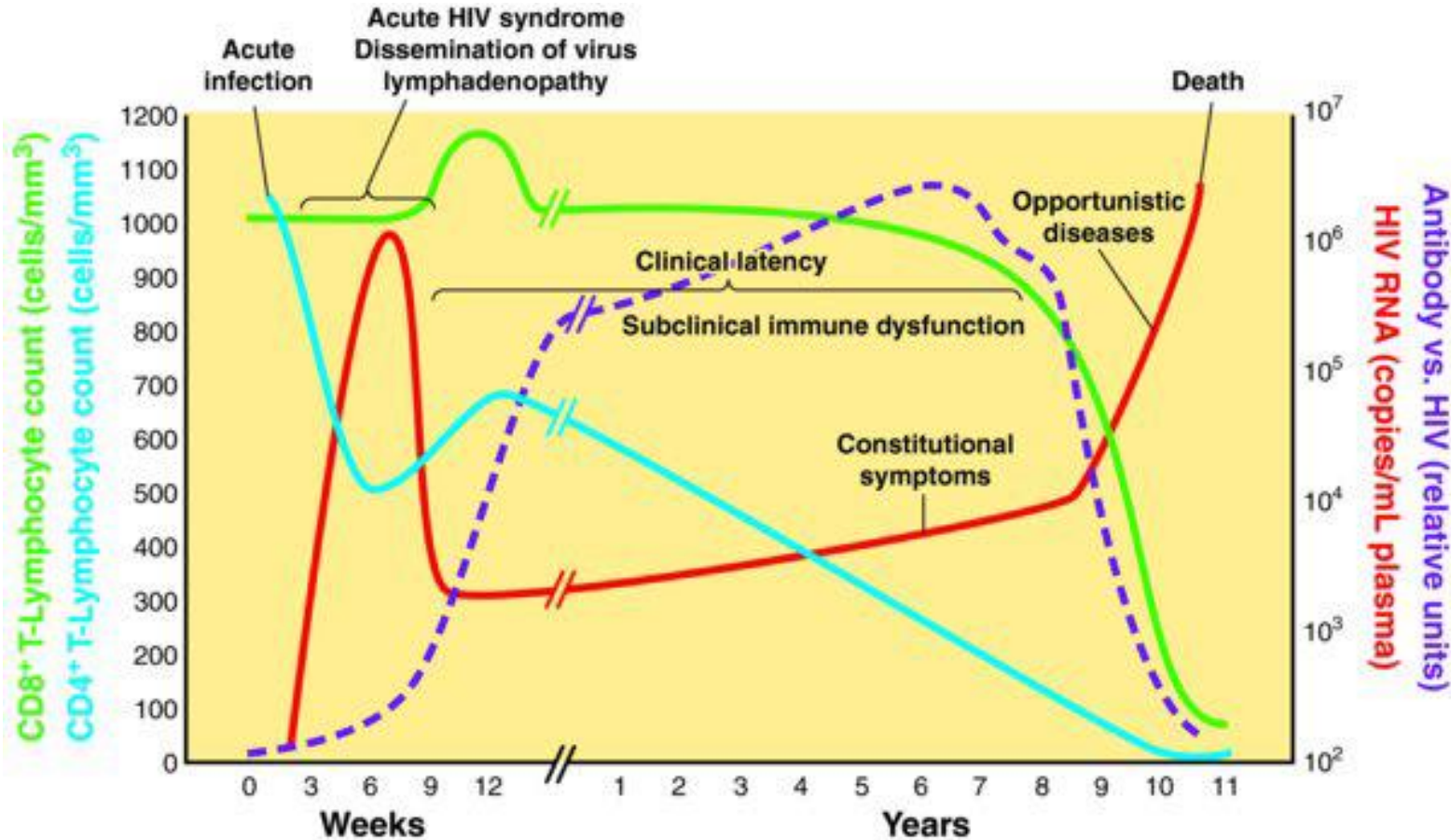
In persons living with untreated HIV infection, at what CD4 count does humoral and cell-mediated immunity precipitously fail?

1. 50
2. 100
3. 200
4. 500

Time Course of Untreated HIV Infection

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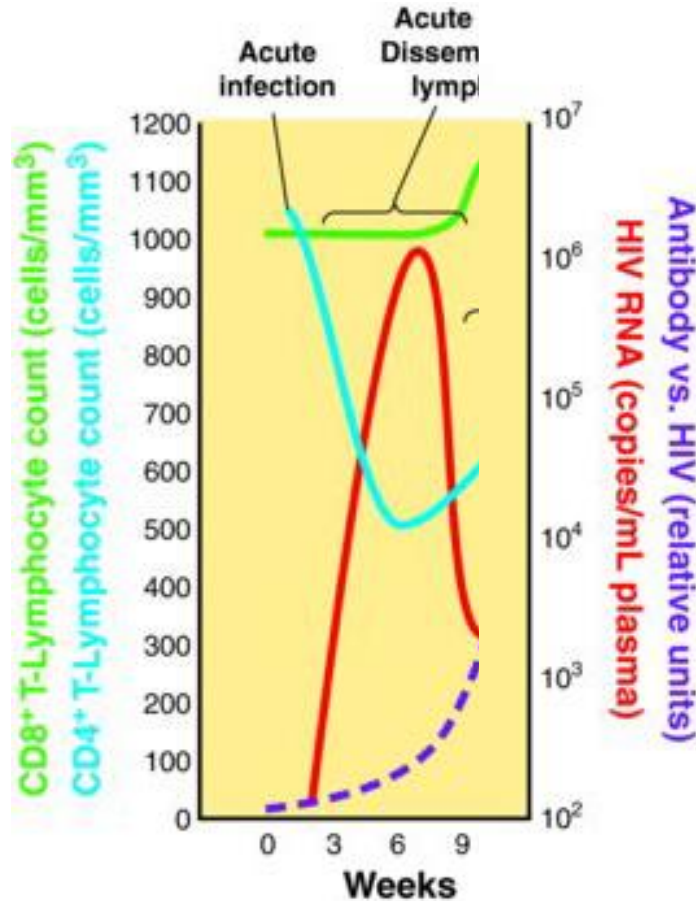


Question 13

Symptoms of the acute retroviral syndrome include all of the below except

1. icterus
2. fever
3. adenopathy
4. oral or vaginal candidiasis

Acute Retroviral Syndrome



- Abrupt onset of fever 2–4 weeks after exposure.
- Accompanied by
 - nonexudative pharyngitis and lymphadenitis,
 - maculopapular skin rash on the head, neck, and upper torso,
 - headache and aseptic meningitis,
 - anorexia, nausea, and diarrhea.
- The acute illness lasts 2–6 weeks. Lethargy and fatigue can persist for several months.

Question 14

Persons living with latent HIV infection are at risk for all the following except

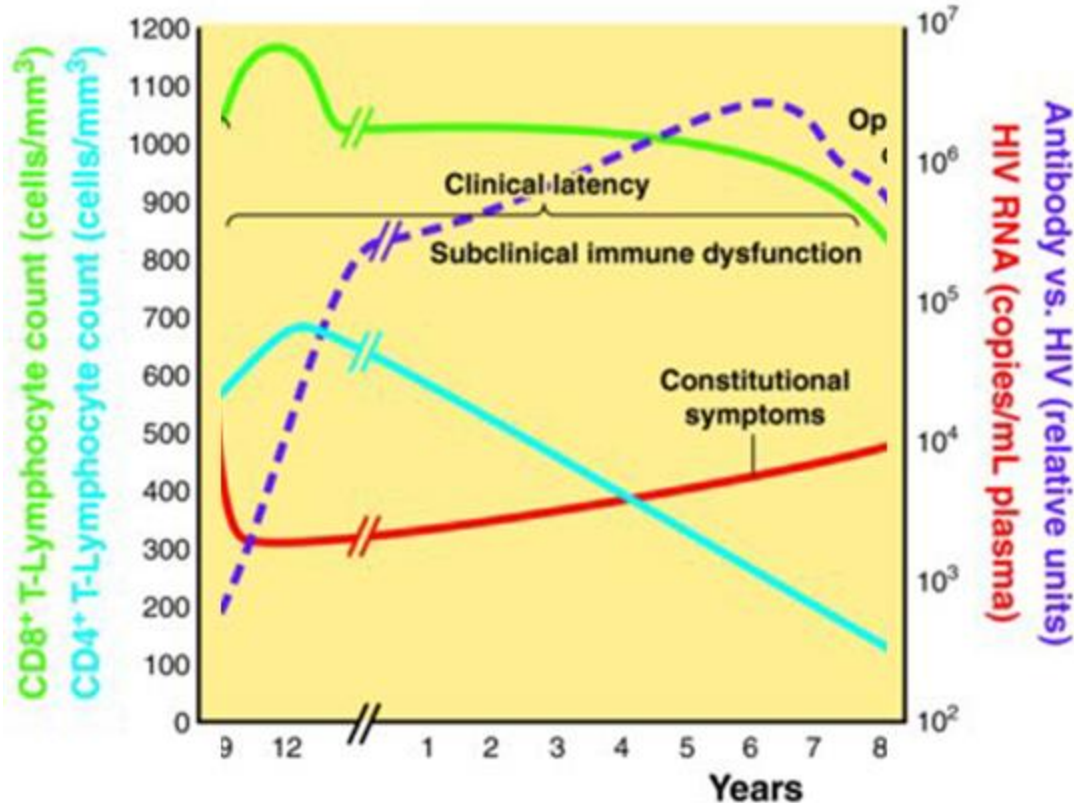
1. premature cardiovascular disease
2. fatigue
3. cytomegalovirus retinitis
4. transmitting HIV to others

Latent HIV infection

Asymptomatic,
transmissible
HIV infection

Progressive risk of
CV disease
Renal disease
Metabolic syndrome
Hyperlipidemia
Fat redistribution
Metabolic acidosis

“Immune lacunae”



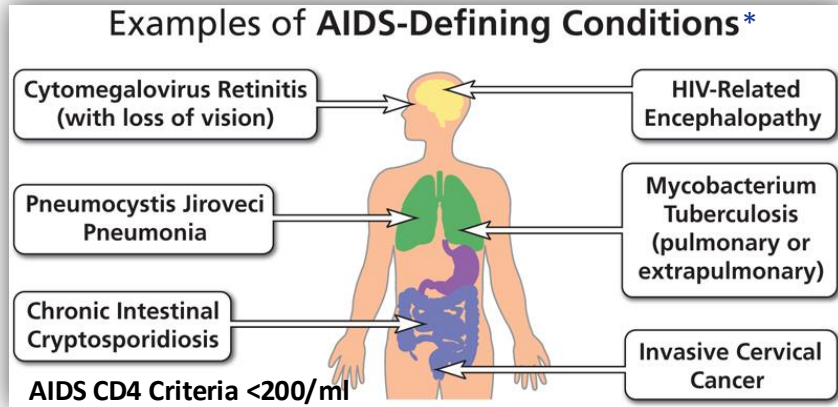
1 in 7 persons with HIV are unaware of their infection

Question 15

All of the following are AIDS-defining illness except:

1. invasive cervical cancer
2. HIV-associated wasting syndrome
3. chronic cryptosporidiosis
4. Hodgkin lymphoma

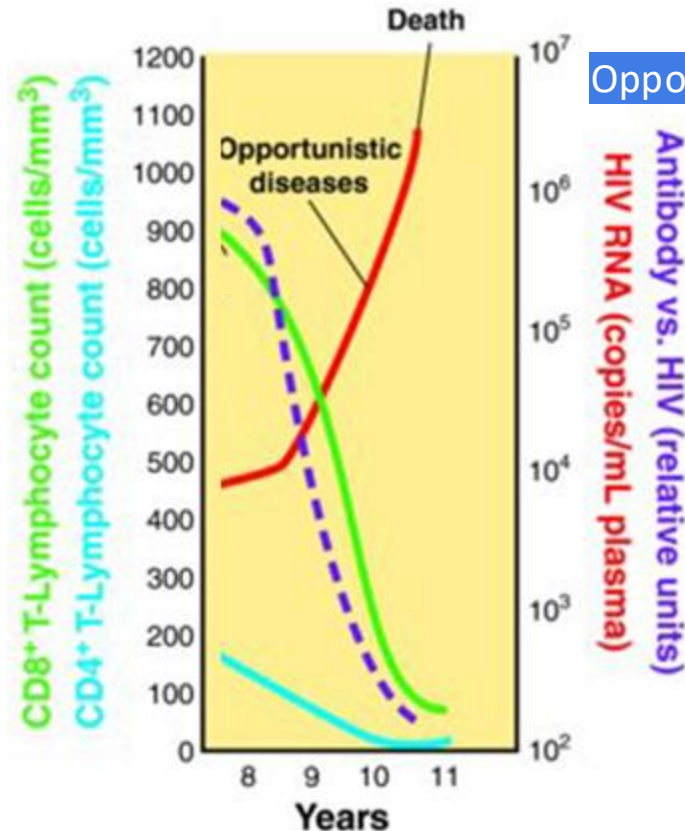
Symptomatic HIV



Opportunistic Infections

CD4		Prophylaxis
<200	<i>Pneumocystis jirovecii</i> *	TMP/SMZ, pentamidine, dapsone
<100	Cerebral toxoplasmosis *	TMP/SMZ
<50	CMV retinitis *	Valganciclovir
	Cryptococcal meningitis *	Fluconazole
	Esophageal candidiasis *	Fluconazole
	<i>M. avium/intracellulare</i>	Azithromycin

*



Opportunistic Cancers

CNS lymphoma*

Kaposi's sarcoma*

Cervical ca*

Hodgkin lymphoma

oral cavity/pharynx

Anus

Liver

Lung

[List of
AIDS-defining
illnesses](#)

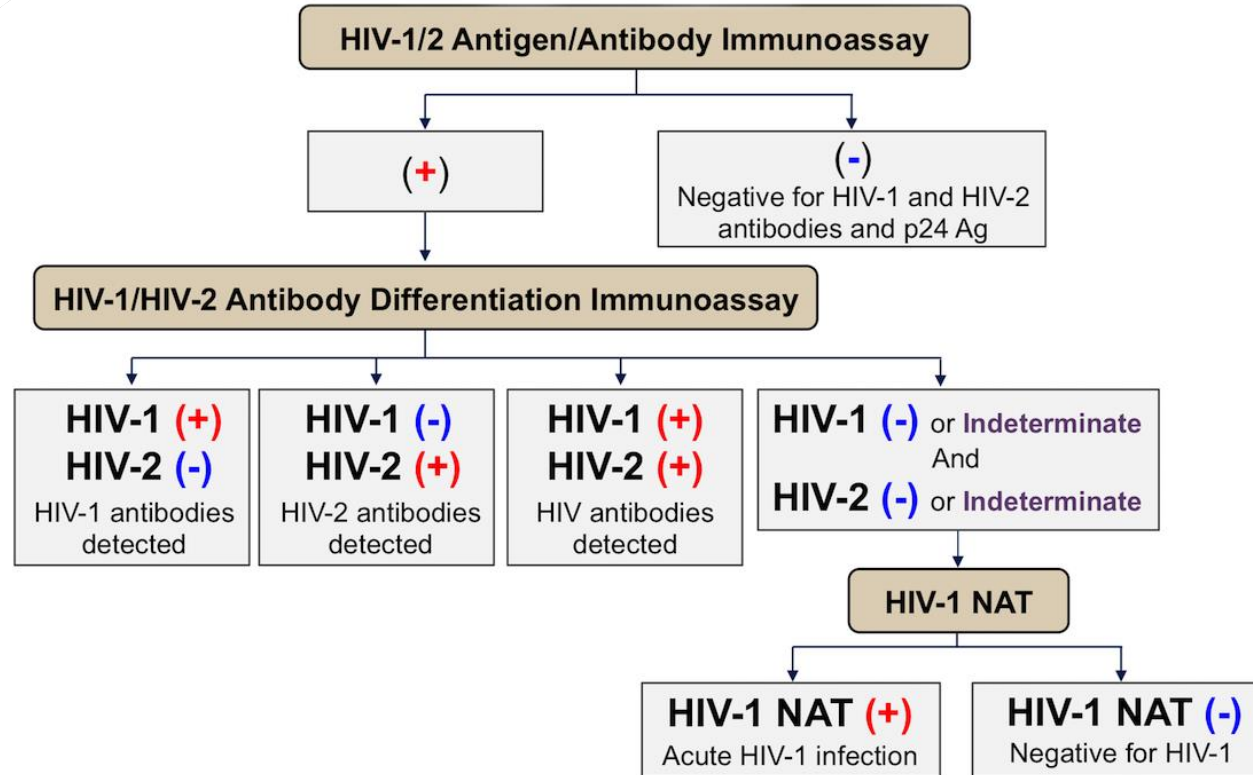
- What HIV test do you offer Bobby?
 - HIV 4th generation antibody-p24 antigen, which is positive. He is understandably upset. You counsel him, recommend 100% barrier protection during sex, and let him know that with behavior modification and treatment he could expect to live to old age.
- What other tests should you order?

Question 16

Persons in whom HIV-1 infection is highly suspected but 4th gen antibody/antigen testing is negative should be tested by

1. Repeat 4th gen ab/ag
2. HIV-1 viral load
3. CD4 count
4. No further testing

HIV 4th Generation Antigen/Antibody Assay



Tests in Persons Newly Diagnosed with HIV

- CD4 cell count, HIV viral load
- HIV-genotype, integrase genotype, coreceptor tropism
- STIs: gonorrhea/chlamydia (TMA), trichomonas (PCR), syphilis (RPR)
- Tuberculosis screening (skin test, IGRA)
- Hepatitis A (total antibody),
Hepatitis B (surface antigen, surface antibody, core total ab),
Hepatitis C (antibody)
- CD4 <100: CMV IgG, Toxoplasma IgG

- Bobby's tests show CD4 is 150/ml and HIV viral load is 150,000/ml. HIV-resistance testing is negative (susceptible to all ART). Hepatitis B surface antibody is reactive and all other testing was negative.
- Should Bobby start treatment and if so, why now and with what medications?

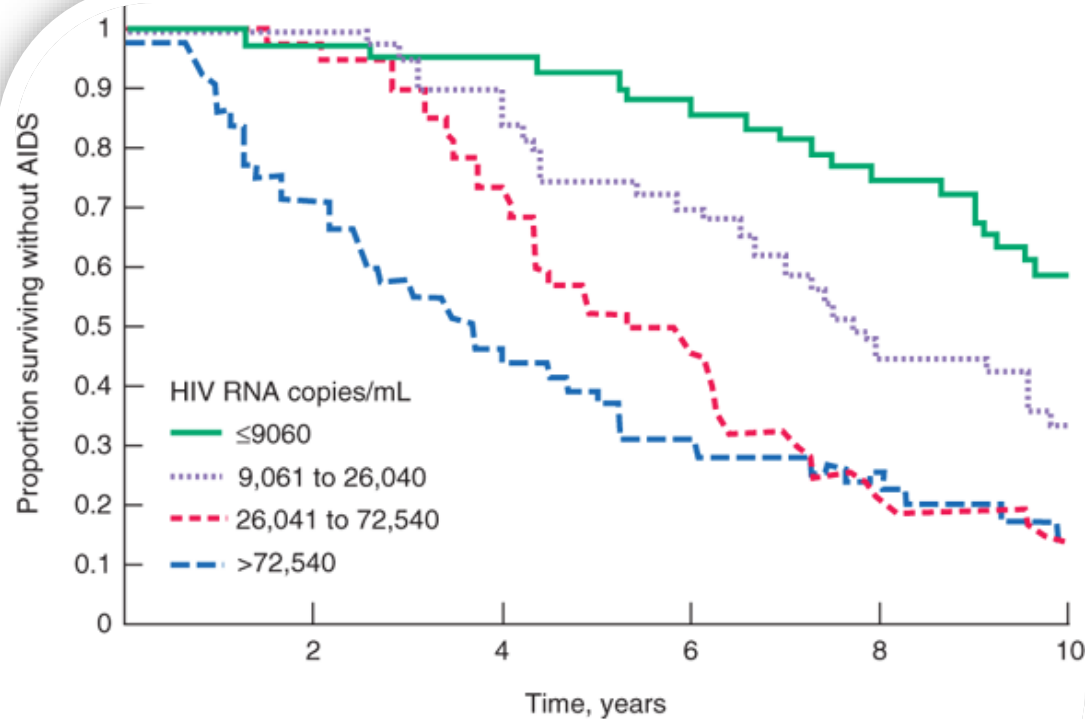
Question 17

In persons treated for HIV, the viral load is predictive of

1. risk of mortality
2. risk of opportunistic infection
3. risk of malignancy
4. ART resistance

HIV Viral Load and Disease Progression

- HIV viral load relation to risk of death
 - 60% 5y mortality if >72540 virions/ml
 - $<5\%$ if <9060
 - Effective ART, undetectable viral load = near-normal lifespan



Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: Harrison's Principles of Internal Medicine, 20th Edition
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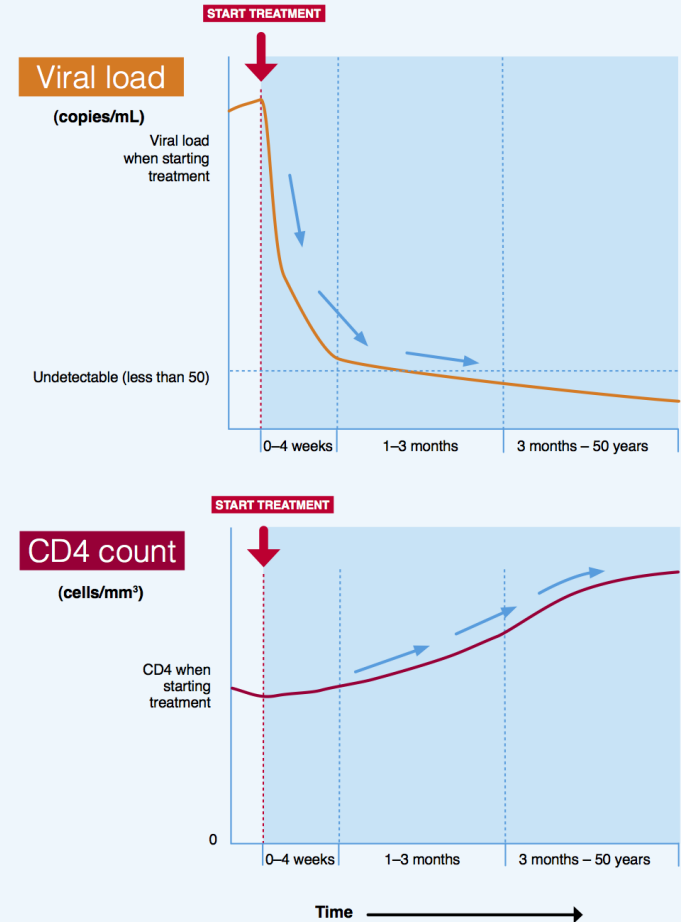
Discussion 3

What document that informs practitioners how to manage persons living with HIV did I suggest you not read?

Bobby

- Bobby begins bictegravir, emtricitabine & tenofovir alafenamide (fixed combination) daily
 - Counsel 100% compliance with ART and safe sex
 - TMP/SMZ DS 3x weekly for PJP prophylaxis
 - Pneumococcal, tetanus, meningococcal and HPV vaccinations

HIV after starting ART



Bobby

- Bobby returns 1 month later. He had no side effects and has been compliant (he set an alarm on his phone as a reminder).
 - CD4 is 220/ml and viral load is 1500/ml.
 - You explain that he can expect an undetectable viral load about 6-8 weeks after starting ART.
- He feels great but explains that he hates taking pills everyday, asks if he needs to do this for the rest of his life and will there ever be a cure. What do you tell him?

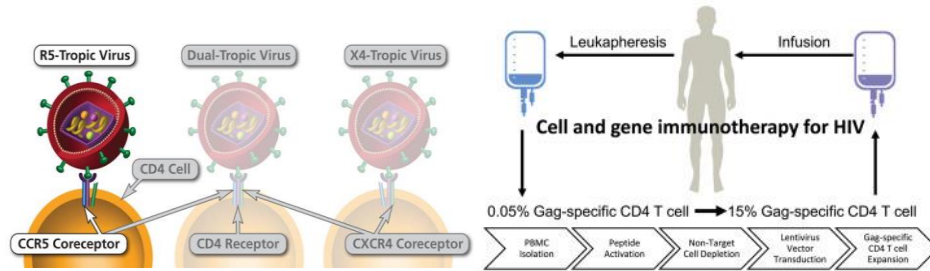
HIV Advances in Therapy and Road to Cure



The NEW ENGLAND
JOURNAL of MEDICINE

Long-Acting Cabotegravir and Rilpivirine for Maintenance of HIV-1 Suppression

Susan Swindells, M.B., B.S., Jaime-Federico Andrade-Villanueva, M.D., Gary J. Richmond, M.D., Giuliano Rizzardini, M.D., Axel Baumgarten, M.D., Mar Masía, M.D., Gulam Latiff, M.D., Vadim Pokrovsky, M.D., Fritz Bredeek, M.D., Graham Smith, M.D., Pedro Cahn, M.D., Yeon-Sook Kim, M.D., Ph.D., et al.



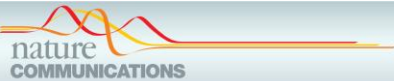
Molecular Therapy
Methods & Clinical Development
Original Article



Preclinical Development and Clinical-Scale Manufacturing of HIV Gag-Specific, Lentivirus Modified CD4 T Cells for HIV Functional Cure

Haishan Li,¹ Tyler Lahusen,¹ Lingzhi Xiao,¹ Nidal Muvarak,¹ Jana Blazkova,² Tae-Wook Chun,² and C. David Pauza¹

HIV Vaccine Remains Elusive



Vaccine elicitation of HIV broadly neutralizing antibodies from engineered B cells

Deli Huang¹, Jenny Tuyet Tran¹, Alex Olson¹, Thomas Vollbrecht², Mary Tenuta¹, Mariia V. Guryleva^{3,4}, Roberta P. Fuller^{1,5,6}, Torben Schiffner^{1,5,6}, Justin R. Abadejos¹, Lauren Couvrette^{1,7}, Tanya R. Blane¹, Karen Saye^{1,5,6}, Wenjuan Li¹, Elise Landais^{1,8}, Alicia Gonzalez-Martin⁹, William Schief^{1,5,6,9}, Ben Mur
Dennis R. Burton^{1,5,6,9,10}, David Nemazee^{1,10} & James E. Voss^{1,10}

The human naive B cell repertoire contains distinct subclasses for a germline-targeting HIV-1 vaccine immunogen

Colin Havenar-Daughton^{1,2,*}, Anita Sarkar^{2,3,4}, Daniel W. Kulp^{2,3,5,6}, Laura Toy^{1,2}, ...
+ See all authors and affiliations

Science Translational Medicine 04 Jul 2018:
Vol. 10, Issue 448, eaat0381
DOI: 10.1126/scitranslmed.aat0381

[Link to video of HIV virion variability](#)

Bobby

- You tell Bobby that there is no cure, but as long as he is compliant and HIV viral load remains undetectable in the future there is the possibility for once-a-month cabotegravir/rilpivirine.
 - He stops TMP/SMZ after CD4 >200 and viral load is undetectable for 6 months on ART.
- 1 year later, he is in a monogamous, discordant relationship and asks how to prevent HIV infection in his partner. What do you tell him?

Question 18

Which of the following will help end the HIV pandemic?

1. PrEP
2. PEP
3. ART
4. USPSTF recommendations for HIV testing

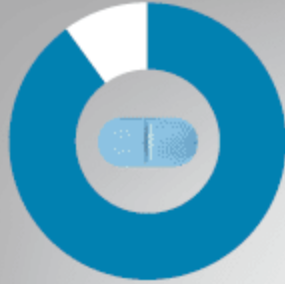
Pre-exposure Prophylaxis (PrEP)



NEW YORK INSTITUTE
OF TECHNOLOGY

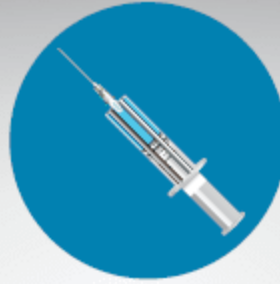
College of Osteopathic
Medicine

Not enough health care providers know about PrEP.



90%

Daily PrEP can reduce the risk of sexually acquired HIV by more than 90%.



70%

Daily PrEP can reduce the risk of HIV infection among people who inject drugs by more than 70%.



1 in 3

1 in 3 primary care doctors and nurses haven't heard about PrEP.

SOURCE: CDC Vital Signs, Dec. 2015.

HIV: Summary

- HIV originated in Africa due to encroachment of forested land exposed humans to SIV.
- HIV is transmitted via blood, sexual and prenatal exposures, all of which can be prevented.
- The viral life cycle can be interrupted at many points by ART to achieve an undetectable viral load.
- CD4 depletion results in progressive and eventually severe immune suppression increasing the risk of infection and malignancy.
- The HIV pandemic has resulted in >40 million deaths and remains active but there are many reasons to be hopeful that it will end during out lifetime.

- cabrah04@nyit.edu
- calendly.com/drcarljabraham
- Office hours: Thursday 1-2 pm CST (2-3 pm EST)
This week: Review questions
- Lecture Feedback Form on Canvas:
- Wilson Hall: Office 424C
- 870-680-8827

Bobby

- Bobby does very well.
- At 5 years CD4 is 775/ml and viral load has remained undetectable.
- He has not missed a single dose of ART.
- His partner is on PrEP and is uninfected.

Bobby

- Bobby misses the next appointment.
 - Concerned, you call him, but his number is not in service and neither is his emergency contact's.
 - A letter comes back “not at this address.”

Bobby

- Bobby is lost to follow up...

Bobby

...until the lecture on opportunistic infections.